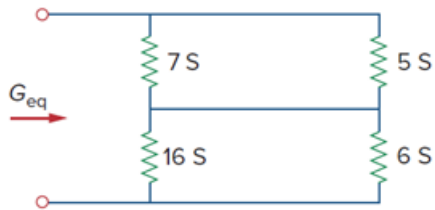


Problem

Calculate G_{eq} in the circuit of Fig. 2.41.

Figure 2.41:



Step-by-step solution

Step 1 of 2

Refer to circuit diagram in Figure 2.41 in the textbook.

The resistors 7 S and 5 S are in parallel.

So their conductance is,

$$\begin{aligned} G_{eq1} &= 7\text{ S} + 5\text{ S} \\ &= 12\text{ S} \end{aligned}$$

Similarly, resistors 16 S and 6 S are in parallel.

Their conductance is,

$$\begin{aligned} G_{eq2} &= 16\text{ S} + 6\text{ S} \\ &= 22\text{ S} \end{aligned}$$

[Comments \(1\)](#)



Anonymous

The Diagram is incorrect, should be 8s. That's why the answer below is not exactly 8 as the textbook indicates.

Step 2 of 2

Calculate the resultant resistors 12 S and 22 S are in series.

$$\begin{aligned} G_{eq} &= G_{eq1} \parallel G_{eq2} \\ &= \frac{(12\text{ S})(22\text{ S})}{(12\text{ S}) + (22\text{ S})} \\ &= \frac{132\text{ S}}{15} \\ &= 7.76\text{ S} \end{aligned}$$

$$\approx 8\text{ S}$$

Therefore, the equivalent conductance, G_{eq} is $\boxed{8\text{ S}}$.